

National University of Computer and Emerging Sciences

Lahore Campus

Operating Systems Lab

Date: 13, 04, 2026

Sessional-I Exam

Total Time (H:M): 1:15

Instructor

Ms Halima Sadia

Semester: Spring-2026

Total Marks: 50

Total Questions: 2

Roll No _____ Section _____ Student Signature _____

Submission path: \\Exam\Lab Submissions\Halima Sadia\OS BCS 4F\Mid 1

Q. No 1: [20 Marks]

Write a C program that demonstrates communication between two processes using **unnamed pipes**.

The program should:

- Accept a **command** (e.g., ls, date, whoami) using **command line arguments**
- The **parent process** creates a pipe and forks a child
- The **child process** executes the given command using **exec()**
- The **output of the command** should be sent through the pipe
- The **parent process** reads from the pipe and displays the output on screen

Ensure proper use of:

- pipe(), fork(), exec(), read(), write()
- Proper closing of unused pipe ends

Q. No. 2: [30 Marks]

Write a C program that uses **multithreading** and **named pipes** to process file data.

The program should:

- Accept a **file name via command line**
- Create a **named pipe (FIFO)**

Thread 1 (Producer):

- Read lines from file
- Send only those lines **having more than 3 words** into FIFO

Thread 2 (Consumer):

- Read lines from FIFO
- Reverse each word in the line
(Example: Hello World → olleH dlroW)
- Print processed lines

Requirements:

- Use pthread and FIFO system calls
- Proper synchronization
- Handle edge cases (empty lines, short lines)